

DATE:30/07/2026

JAVA COLLECTIONS AND FRAMEWORK

1.Logger:

The screenshot shows the SIMATS Online Programming Lab interface. The top header includes the SIMATS Engineering logo, the lab name, and a user profile for MULLA MAHAMMAD RAFI. The left sidebar displays 'QUESTIONS: 8 / 8 completed' and a dropdown menu for 'Logger'. The main 'PROGRAM EDITOR' area contains the following Java code:

```
1 interface logger {
2     void debug();
3     void info();
4 }
5 class Log implements logger {
6     public void debug() {
7         System.out.println("Debug Message");
8     }
9     public void info() {
10        System.out.println("Info Message");
11    }
12    public static void main(String[] args) {
13        Log l = new Log();
14        l.debug();
15        l.info();
16    }
17 }
```

Below the editor is the 'OUTPUT' section, which displays:

```
Debug Message
Info Message
```

The right sidebar has an 'INPUT' section with the text 'Your INPUT goes here!'.

2.Character:

The screenshot shows the SIMATS Online Programming Lab interface. The top header includes the SIMATS Engineering logo, the lab name, and a user profile for MULLA MAHAMMAD RAFI. The left sidebar displays 'QUESTIONS: 8 / 8 completed' and a dropdown menu for 'Character'. The main 'PROGRAM EDITOR' area contains the following Java code:

```
1 interface Character {
2     void move();
3     void attack();
4     void damage();
5 }
6 class Hero implements Character {
7     public void move() {
8         System.out.println("Hero moves");
9     }
10    public void attack() {
11        System.out.println("Hero attacks");
12    }
13    public void damage() {
14        System.out.println("Hero takes damage");
15    }
16    public static void main(String[] args) {
17        Hero h = new Hero();
18        h.move();
19        h.attack();
20        h.damage();
21    }
22 }
```

Below the editor is the 'OUTPUT' section, which displays:

```
Hero moves
Hero attacks
Hero takes damage
```

The right sidebar has an 'INPUT' section with the text 'Your INPUT goes here!'.

3.Sort:



SIMATS Online Programming Lab
JAVA PROGRAMMING

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QUESTIONS 8 / 8 completed

Character

CHARACTER

You are developing a game that involves different types of characters, such as heroes, villains, and monsters. Write a Java program to create an interface called "Character" with methods for moving, attacking, and taking damage. Implement the interface in classes for each type of character, such as "Hero", "Villain" and

PROGRAM EDITOR

Java

Run

Save

```
1 interface Character {
2     void move();
3     void attack();
4     void damage();
5 }
6 class Hero implements Character {
7     public void move() {
8         System.out.println("Hero moves");
9     }
10    public void attack() {
11        System.out.println("Hero attacks");
12    }
13    public void damage() {
14        System.out.println("Hero takes damage");
15    }
16    public static void main(String[] args) {
17        Hero h = new Hero();
18        h.move();
19        h.attack();
20        h.damage();
21    }
22 }
```

OUTPUT

Hero moves
Hero attacks
Hero takes damage

INPUT

Your INPUT goes here!

4.Hashset:



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QUESTIONS 8 / 8 completed

HashSet

HASHSET

Write a Java program to create a HashSet of strings and remove a specific element from it.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.HashSet;
2 public class Solution {
3     public static void main(String[] args) {
4         HashSet<String> set = new HashSet<>();
5         set.add("Apple");
6         set.add("Banana");
7         set.add("Orange");
8         set.add("Mango");
9         System.out.println(set);
10    }
11 }
```

OUTPUT

[Apple, Mango, Orange, Banana]

INPUT

Apple

5.TreeMap:

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the lab name, and the user's name (MULLA MAHAMMAD RAFI) and ID (192421364RD). The main content area is divided into three sections: QUESTIONS, PROGRAM EDITOR, and INPUT.

QUESTIONS (8 / 8 completed):

- TreeMap** (selected)
- TREEMAP**
- Write a Java program to create a TreeMap of strings and integers and iterate over its entries.

PROGRAM EDITOR (Language: Java):

```
1 import java.util.*;
2 public class TreeMapDemo {
3     public static void main(String[] args) {
4         TreeMap<String, Integer> map = new TreeMap<>();
5         map.put("Ram", 90);
6         map.put("John", 85);
7         map.put("Ravi", 95);
8         for (Map.Entry<String, Integer> e : map.entrySet()) {
9             System.out.println(e.getKey() + " " + e.getValue());
10        }
11    }
12 }
```

OUTPUT:

```
John 85
Ram 90
Ravi 95
```

INPUT: Your INPUT goes here!

6.PriorityQueue:

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the lab name, and the user's name (MULLA MAHAMMAD RAFI) and ID (192421364RD). The main content area is divided into three sections: QUESTIONS, PROGRAM EDITOR, and INPUT.

QUESTIONS (8 / 8 completed):

- PriorityQueue** (selected)
- PRIORITYQUEUE**
- Write a Java program to create a PriorityQueue of integers and add elements to it.

PROGRAM EDITOR (Language: Java):

```
1 import java.util.PriorityQueue;
2 public class Main {
3     public static void main(String[] args) {
4         PriorityQueue<Integer> pq = new PriorityQueue<>();
5         pq.add(30);
6         pq.add(10);
7         pq.add(20);
8         System.out.println(pq);
9     }
10 }
```

OUTPUT:

```
[10, 30, 20]
```

INPUT: Your INPUT goes here!

7. LinkedHash Map:

The screenshot shows the SIMATS Online Programming Lab interface. The top header includes the SIMATS Engineering logo, the lab name, and a user profile. The left sidebar shows a list of questions, with 'LINKEDHASHMAP' selected. The main area contains a 'PROGRAM EDITOR' with a Java code snippet that creates a LinkedHashMap and adds three entries. Below the editor is an 'OUTPUT' section showing the result. To the right is an 'INPUT' section.

QUESTIONS 8 / 8 completed

LinkedHashMap

LINKEDHASHMAP

Write a Java program to create a LinkedHashMap of strings and integers and retrieve the keys in the order they were inserted.

PROGRAM EDITOR Java Run Save

```
1 import java.util.LinkedHashMap;
2 public class Main {
3     public static void main(String[] args) {
4         LinkedHashMap<String, Integer> map = new LinkedHashMap<>();
5         map.put("Apple", 10);
6         map.put("Banana", 20);
7         map.put("Orange", 30);
8         System.out.println(map.keySet());
9     }
10 }
```

OUTPUT

```
[Apple, Banana, Orange]
```

INPUT

Your INPUT goes here!

8. Max-Min:

The screenshot shows the SIMATS Online Programming Lab interface. The top header is the same as the previous one. The left sidebar shows a list of questions, with 'MAX-MIN' selected. The main area contains a 'PROGRAM EDITOR' with a Java code snippet that creates a HashSet, adds five integers, and prints the set, maximum, and minimum values. Below the editor is an 'OUTPUT' section showing the result. To the right is an 'INPUT' section.

QUESTIONS 8 / 8 completed

Max-Min

MAX-MIN

Write a Java program to create a Set of integers and perform different operations on it, such as finding the maximum and minimum values.

PROGRAM EDITOR Java Run Save

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Set<Integer> set = new HashSet<>();
5         set.add(20);
6         set.add(50);
7         set.add(10);
8         set.add(40);
9         System.out.println("Set: " + set);
10        System.out.println("Maximum: " + Collections.max(set));
11        System.out.println("Minimum: " + Collections.min(set));
12    }
13 }
```

OUTPUT

```
Set: [50, 20, 40, 10]
Maximum: 50
Minimum: 10
```

INPUT

Your INPUT goes here!